

IAN REED SHEHADEH

Baltimore, Maryland, US 21231

ian@shehadeh.net ♦ ian.shehadeh.net ♦ linkedin.com/in/irshehadeh/ ♦ github.com/ishehadeh

SUMMARY

Software engineer with a background in programming language design and production fintech systems. Led development through team turnover, kept billing jobs running in production, wrote documentation, and onboarded three new developers.

WORK EXPERIENCE

The Land Geek (GeekPay)

Remote

Software Engineer

Jan 2025 – Present

- Migrated billing and loan status jobs from Rails to Laravel, serving 11,000+ loans and charging \$2.5M+/mo.
- added dry-run modes and rate limiting to support staged production rollouts.
- Shipped targeted fixes that cut monthly support ticket volume from ~250 to ~80.
- Became sole engineer through team turnover; owned production releases, resolved live incidents.
- Onboarded three new developers; established coding standards through systematic code review.
- Created banner notification with new PostgreSQL tables, controllers, and UI (Tailwind, Blade, Alpine.js).

Under Armour

Remote

Software Engineering Intern

May 2024 – Aug 2024

- Added structured detail to Firebase crash logs, helping the team resolve of recurring production crashes.
- Built screenshot share notifications, coordinating requirements across product, design, and QA.

PROJECTS

cgajs

<https://github.com/ishehadeh/cgajs>

A library for solving and analyzing combinatorial games in TypeScript

Feb 2025 – present

- BigInt-backed bit boards for unbounded board size and exact positions in JS.
- Canonical-form rewriting to normalize and compare arbitrary game trees.
- Matches reference values for the game Konane.

Howlite

<https://github.com/ishehadeh/howlite>

WIP language targeting RISC-V (parser, type checker, no codegen)

Nov 2023 – Dec 2024

- Built Rust parser and type checker; validated integer-range types and structural subtyping.
- Type narrowing in conditionals so control flow refines integer types.
- Expresses integer types as a set of continuous ranges, which can be used in array indexing and arithmetic.
- Type-checks generic functions with range-bounded array-length parameters and union return types.
- Concluded that structural typing scales better than parametric polymorphism for a low-level language.

Janet (open-source contributions)

<https://github.com/janet-lang/janet>

Contributions to the Janet programming language runtime (C)

Jan 2022 – Feb 2022

- Fixed cross-platform async I/O bugs in Linux (epoll) and Windows (overlapped I/O); merged upstream.
- Implemented rest-pattern destructuring for the match macro, with nested pattern support.

EDUCATION

St. Mary's College of Maryland

St. Mary's City, Maryland

Bachelor of Science in Computer Science and Mathematics

Aug 2020 – May 2025

- **Courses:** Algorithms & Data Structures, Theory of Computation, Analysis, Abstract Algebra I & II

SKILLS

- **Languages:** Rust, PHP, TypeScript
- **Backend:** Laravel, PostgreSQL
- **Frontend:** Tailwind CSS, Laravel Blade, Alpine.js, HTML, CSS
- **Tools:** GitHub Actions, Jujutsu VCS, Docker